

Notes: Ghosh, Vall and Zeng (JF 2025)
FinTech Lending and Cashless Payments

Contents

1	Executive Takeaways	3
2	Research question, empirical object, and contribution map	3
2.1	Core question	3
2.2	Main empirical objects (what is measured)	3
2.3	What is conceptually new relative to standard credit-screening papers	3
3	Data and construction of key variables	4
3.1	Data source and sample	4
3.2	Summary statistics (Table I: what the sample looks like)	4
3.3	Classifying payment technologies (conceptual steps)	4
3.4	Main regressor	4
4	Empirical strategy: the regression models (“all the empirical models”)	5
4.1	Baseline OLS for financing outcomes (Equation (1))	5
4.2	Default and repayment performance (Equation (2))	5
4.3	Within-borrower design for repeat applicants (Equation (3))	5
4.4	Complementarity with other quality signals (Equation (4))	5
4.5	Instrumental variables design (2SLS, Table VIII)	6
5	Main empirical findings: interpreting each figure and table	6
5.1	Figure 1: binned scatter evidence for financing outcomes	6
5.2	Table II: OLS financing outcomes (the “workhorse” result)	6
5.3	Figure 2 and Table III: default, time to delinquency, and repayment	7
5.4	Table IV: other transaction records (online marketplace data)	8
5.5	Table V: informational content (inflows vs. outflows; information-intensive payments)	8
5.6	Table VI: repeat applicants and borrower fixed effects (within-firm evidence)	9
5.7	Table VII: cashless usage complements other quality signals	9
5.8	Figure 3 and Table VIII: IV evidence from demonetization-related variation	10
6	Theory model (Section VI): “all the theoretical models” + core derivations	11
6.1	Setup and timeline	11
6.2	Types and diversion risk	11
6.3	Cash vs. cashless as a verifiability/precision choice (Equation (5))	11
6.4	Default cost for cashless borrowers (Equation (6))	12
6.5	Borrower utilities (Equations (7)–(8))	12

6.6	Expected default cost (Equation (13)) — derivation sketch	12
6.7	Competitive pricing and break-even (implied loan rates)	12
6.8	Incentive compatibility and individual rationality (Equations (9)–(12))	13
6.9	Proposition 1 (selection effect): existence of separating equilibrium	13
6.10	Proposition 2 (accuracy effect): complementarity with borrower quality	14
6.11	Appendix C: alternative information structure (robustness of the model)	14

1 Executive Takeaways

This paper documents a tight *informational synergy* between *cashless payment usage* and *FinTech lending*. In micro and small business unsecured lending in India (Indifi, 2015–2022), firms with a higher share of cashless payments prior to application obtain better financing outcomes (higher approval, lower offered interest rate, larger loan amount) and subsequently default less. These patterns are stronger when cashless technologies generate *more precise* and *more granular* transaction traces (e.g. information-intensive records), and when the cashless behavior is measured on *outflows* rather than inflows.

2 Research question, empirical object, and contribution map

2.1 Core question

Why do FinTech lenders (who lack traditional relationship data and deposit histories) succeed in screening borrowers, and how does payment technology adoption (cash vs. cashless) reshape credit access and default?

2.2 Main empirical objects (what is measured)

- **Payment technology usage (pre-application):** a value-weighted share of transactions executed via cashless methods (overall and by inflows/outflows), plus a decomposition into *information-intensive* vs. *information-light* cashless technologies.
- **Financing outcomes:** (i) approval indicator; (ii) offered interest rate; (iii) offered loan amount (log); and additional performance outcomes (default, time to delinquency, time to repayment).

2.3 What is conceptually new relative to standard credit-screening papers

The key innovation is to treat *cashless payment usage itself* as an economically meaningful and measurable signal/commitment device that can be exploited by FinTech lenders (especially under open banking). The paper distinguishes:

1. **Information content/precision:** some cashless technologies produce more precise verifiable traces (higher “*q*” in the model).
2. **Directional content:** cashless *outflows* are more informative than cashless inflows.
3. **Complementarity:** cashless usage works *with*, not *instead of*, other borrower-quality signals.

3 Data and construction of key variables

3.1 Data source and sample

The empirical analysis uses proprietary data from **Indifi**, a large Indian FinTech lender, covering **unsecured** loans to micro and small businesses from **September 2015 to September 2022**. Applications are matched to (i) loan screening decisions and contract terms, (ii) ex-post loan performance, and crucially (iii) transaction-level **bank-statement payment records** for the **six months prior** to application.

3.2 Summary statistics (Table I: what the sample looks like)

Table I is the baseline reality check:

- **Scale:** 316,719 complete applications; 152 million transactions.
- **Borrowers:** mean owner age ≈ 36.6 , business age ≈ 4.3 years.
- **Credit bureau:** mean Cibil score ≈ 693 (with 11.6% missing).
- **Payment usage:** average share cashless ≈ 0.779 ; average share cash ≈ 0.221 .
- **Financing outcomes:** approval rate ≈ 0.208 ; among offered loans, mean interest rate $\approx 25.4\%$ and average amount $\approx 535\text{k INR}$.
- **Default:** among loans with performance data, default rate $\approx 17.5\%$.

These magnitudes matter because the results are economically meaningful only if the baseline default risk and borrowing costs are nontrivial.

3.3 Classifying payment technologies (conceptual steps)

Using the text labels in bank statements, transactions are classified into:

1. **Cash payments:** cash deposits/withdrawals (ATM, branch, etc.).
2. **Cashless payments:** electronic transfers, card/UPI-type payments, etc.
3. **Information-intensive vs. information-light cashless:**
 - *Information-intensive* records are more disaggregated and better identify counterparties / contexts (richer verifiable signal).
 - *Information-light* records are more aggregated or less identifiable (noisier for screening).

3.4 Main regressor

Let $CashlessShare_i$ be the applicant-level value-weighted share of transactions executed via cashless methods over the six months prior to application. The paper standardizes this variable (so coefficients can be read as the effect of a one-standard-deviation increase in cashless usage).

4 Empirical strategy: the regression models (“all the empirical models”)

4.1 Baseline OLS for financing outcomes (Equation (1))

The baseline specification relates financing outcomes to cashless usage:

$$\text{FinancingOutcome}_i = \beta \text{CashlessShare}_i + \gamma' X_i + \text{FE}_i + \varepsilon_i, \quad (1)$$

where:

- $\text{FinancingOutcome}_i$ is (i) approval indicator, (ii) offered interest rate, or (iii) $\log(\text{offered amount})$;
- X_i includes borrower controls (e.g. $\log \#$ payment records, credit history length, business age/vintage, $\log$ owner age, missing score indicator, top-up indicator);
- FE_i stacks rich fixed effects: industry FE, application month FE, and (in the most demanding columns) Cibil-score-group FE, 3-digit zip FE, and revenue-decile FE.

Standard errors are clustered at the application-month level.

4.2 Default and repayment performance (Equation (2))

Loan performance is modeled similarly:

$$\text{Default}_i = \beta \text{CashlessShare}_i + \gamma' X_i + \text{FE}_i + \varepsilon_i. \quad (2)$$

The paper also augments (2) with additional controls capturing the *intensive* screening margin (offered interest rate and amount) to ask whether cashless usage contains incremental information beyond what the lender already priced.

4.3 Within-borrower design for repeat applicants (Equation (3))

To mitigate time-invariant unobserved heterogeneity (“better firms both use cashless and get approved”), the paper uses repeat applicants (top-up loans) and runs:

$$\text{Approved}_{i,t} = \beta \text{CashlessShare}_{i,t} + \mu_i + \delta_t + \psi \text{Score}_{i,t} + \varepsilon_{i,t}, \quad (3)$$

where μ_i are borrower fixed effects and δ_t are application time fixed effects. Identification comes from within-borrower changes in cashless usage across applications.

4.4 Complementarity with other quality signals (Equation (4))

To test whether cashless usage substitutes for or complements other borrower-quality signals:

$$\text{FinancingOutcome}_i = \beta_1 \text{CashlessShare}_i + \beta_2 Q_i + \beta_3 (\text{CashlessShare}_i \times Q_i) + \gamma' X_i + \text{FE}_i + \varepsilon_i, \quad (4)$$

where Q_i is a proxy for borrower quality (e.g. credit score, or payment-outflow volatility).

- If $\beta_3 > 0$ for favorable outcomes (approval, amount) and $\beta_3 < 0$ for unfavorable outcomes (interest rate, default), then **cashless usage and quality are complements**.
- Intuitively, cashless usage improves the *accuracy* of screening more when a borrower is genuinely high quality (or when combined signals jointly reduce uncertainty).

4.5 Instrumental variables design (2SLS, Table VIII)

To strengthen causal interpretation, the paper instruments for cashless usage using the post-2016 demonetization cash shortages and differences across bank branches:

- **Instrument:** an indicator for banking at a *currency chest* branch interacted with a dummy for the shortage period (2018–2019).
- Economic logic: cash shortages were less intense at currency chest branches, so their customers could rely relatively more on cash and less on cashless payments in the shortage period, shifting *CashlessShare* for plausibly exogenous reasons.

The first stage is:

$$CashlessShare_i = \pi_0 + \pi_1 (\text{ChestBranch}_i \times \mathbf{1}\{\text{Year} \in 2018\text{--}2019\}) + \Gamma' X_i + FE_i + u_i,$$

and the second stage replaces *CashlessShare_i* in (1) by its fitted value.

5 Main empirical findings: interpreting each figure and table

5.1 Figure 1: binned scatter evidence for financing outcomes

Figure 1 provides a visual “first pass”:

- **Panel A: Approval vs. cashless share** is upward sloping: higher cashless usage is associated with higher approval probability.
- **Panel B: Offered interest rate vs. cashless share** is downward sloping: higher cashless usage is associated with lower offered rates.
- **Panel C: Offered amount vs. cashless share** is upward sloping.

The binned scatter is residualized for application year-month fixed effects, so it is not purely time-trend driven.

5.2 Table II: OLS financing outcomes (the “workhorse” result)

Table II estimates (1) for three outcomes.

Approval (columns 1–2). A one-standard-deviation increase in *CashlessShare* raises approval probability by about **1.7 percentage points** (column (2): 0.017). Relative to the baseline approval rate 0.208 (Table I), this is economically meaningful.

Interest rate (columns 3–4). Conditional on approval, a one-standard-deviation increase in *CashlessShare* reduces the offered interest rate by about **44 bps** (column (4): -0.440 percentage points).

Loan amount (columns 5–6). The coefficient on $\log(\text{Amount})$ is about 0.134, meaning roughly a **13–14%** larger offered amount for a one-s.d. increase in cashless usage.

Interpretation. These results are consistent with the idea that cashless usage provides information that improves the lender’s screening and pricing, shifting the borrower along both the extensive margin (approval) and intensive margins (rate and amount).

5.3 Figure 2 and Table III: default, time to delinquency, and repayment

Figure 2 (visual). A binned scatter shows a clear negative relationship between cashless usage and default probability.

Default probability (Table III, columns 1–2). A one-standard-deviation increase in *CashlessShare* reduces default probability by about **2 percentage points** (column (2): approximately -0.023), which is around **11%** of the baseline default rate $\approx 17.5\%$ (Table I).

Adding pricing controls (Table III, columns 3–4). Including $\log(\text{interest rate})$ and $\log(\text{amount})$ attenuates the coefficient but it remains statistically significant. The paper interprets this as cashless usage containing incremental information that the lender did not fully price into contract terms (possibly due to learning frictions).

Intensive performance margins (Table III, columns 5–8). Higher cashless usage predicts:

- **Later delinquency** (more days until first delinquency, conditional on delinquency).
- **Faster repayment** (fewer days until full repayment), interpreted as linked to short-term liquidity needs.

Numbers to remember (Table III). In the preferred specifications with the most comprehensive fixed effects, a one-s.d. increase in cashless usage is associated with roughly **5 additional days** until first delinquency (conditional on delinquency) and around **7–12 fewer days** until full repayment (faster repayment), consistent with cashless usage capturing both creditworthiness and short-term liquidity.

5.4 Table IV: other transaction records (online marketplace data)

Table IV asks whether the logic extends beyond bank-statement cashless payments.

Setup. Some applicants have revenue shares from an online marketplace with a data-sharing agreement. This marketplace-share variable is interpreted as an alternative outside data source for screening.

Result. A higher share of marketplace transactions is associated with:

- higher approval (≈ 0.061),
- lower interest rates (≈ -0.325),
- higher amounts (≈ 0.034 log points),
- lower default (≈ -0.006),

even when controlling for *CashlessShare*. This supports the view that *multiple data sources can be cumulative* in screening value.

5.5 Table V: informational content (inflows vs. outflows; information-intensive payments)

Table V refines *why* cashless usage matters by decomposing it along two dimensions.

Quantitative magnitudes (Table V). Because *CashlessShare* is standardized, coefficients can be compared across decompositions. In Panel A:

- **Approval:** outflows (≈ 0.019 – 0.022) are about 2 – $3\times$ as large as inflows (≈ 0.007 – 0.008).
- **Loan amount (log):** outflows (≈ 0.144 – 0.150) exceed inflows (≈ 0.068 – 0.077).
- **Default:** outflows reduce default by about 2.0 – 2.1 pp, versus 0.9 – 1.1 pp for inflows.

In Panel B, the share of information-intensive payments has even stronger effects: approval rises by about 4.2 – 5.0 pp and offered interest rates fall by about 0.64 – 0.71 percentage points for a one-s.d. increase.

Panel A: inflows vs. outflows. Both cashless inflows and cashless outflows predict better financing outcomes and lower default, but **outflows are quantitatively stronger**. This is consistent with the idea that:

- outflows are more under borrower discretion (a stronger revealed-choice signal),
- outflows more directly map to resource diversion and “willful default” risk.

Panel B: information-intensive cashless payments. The share of information-intensive payments has especially large coefficients:

- stronger approval effects,
- larger interest rate reductions,
- larger amount increases,
- larger default reductions,

which supports the paper’s precision channel: *more informative* records are more valuable for screening.

5.6 Table VI: repeat applicants and borrower fixed effects (within-firm evidence)

Table VI focuses on top-up loans (repeat applicants) and estimates (3).

Main point. Even **within the same borrower**, higher cashless usage predicts higher approval. This reduces the concern that the baseline OLS results are driven solely by cross-sectional unobserved quality (“good firms both go cashless and get approved”).

Numbers to remember (Table VI). With borrower fixed effects (column (2)), a one-s.d. within-borrower increase in cashless usage increases approval by about **0.9 percentage points** (0.009). When splitting cashless into revenue vs. spending (column (3)), the spending/outflow component is the economically important one (≈ 0.010), while the revenue/inflow component is small and statistically weak. Using the information-intensive share (column (4)) yields an even larger within-borrower approval effect (≈ 0.021).

Decomposition. Within-borrower variation suggests that **cashless spending/outflows** and **information-intensive payments** are particularly predictive of approval.

5.7 Table VII: cashless usage complements other quality signals

Table VII implements (4).

Panel A (credit score). Both Cibil score and *CashlessShare* predict better terms. Importantly, interaction terms indicate that the benefit from cashless usage is **stronger for higher-score borrowers** for approval/amount and more favorable pricing, consistent with the paper’s *accuracy effect*.

Numbers to remember (Table VII). Panel A (sample with Cibil score) shows strong main effects for both signals and meaningful interactions:

- **Approval:** cashless (≈ 0.023) and score (≈ 0.050) both increase approval; the interaction is positive (≈ 0.012), meaning cashless is more valuable when score is higher.
- **Pricing:** cashless lowers offered rates (≈ -0.432) and the interaction is strongly negative (≈ -0.219), implying the rate benefit of cashless is larger for higher-score borrowers.
- **Amount:** cashless raises amounts (≈ 0.125) and the interaction is positive (≈ 0.039).

Panel B uses weekly outflow volatility as a (negative) quality proxy. The interaction terms imply that cashless is more beneficial when volatility is *lower* (better firms): for example, the approval interaction is about -0.002 , and the interest-rate interaction is about $+0.155$ (offsetting the negative cashless main effect as volatility rises).

Panel B (payment-based volatility). Using weekly outflow volatility as a payment-record-based risk proxy:

- volatility itself is associated with worse terms (e.g., smaller loan amounts),
- the interaction implies that cashless usage provides **larger benefits when volatility is lower** (i.e., for better firms), again consistent with improved screening accuracy.

5.8 Figure 3 and Table VIII: IV evidence from demonetization-related variation

Figure 3 (first stage visualization). Applicants banking at currency chest branches exhibit a distinct pattern in cashless usage over time, with a sharper relative decline during the shortage years (2018–2019).

Table VIII (2SLS). The instrument is strong (effective F -statistics ≈ 27 to 37). The second stage implies a substantially larger approval effect of cashless usage than OLS:

$$\hat{\beta}^{2SLS} \approx 0.12 \text{ to } 0.15 \quad (\text{approval probability per 1 s.d. increase in } \textit{CashlessShare}).$$

The IV interaction with Cibil score is positive, consistent with complementarity.

Numbers to remember (Table VIII).

- **First stage:** the chest $\times$ 2018–2019 instrument reduces cashless usage by about 0.11–0.13 s.d., with effective F -statistics around 27–37.
- **Second stage:** instrumented cashless usage increases approval by about 0.12–0.15 (12–15 pp). With the interaction, the marginal effect of cashless usage is increasing in Cibil score (interaction coefficient ≈ 0.056).

How I interpret the OLS vs. IV gap. A natural reading is that 2SLS identifies a *local average treatment effect* for compliers whose cashless adoption is shifted by cash-supply frictions. Those compliers may be exactly the margin where cashless records are most valuable for screening, amplifying the implied effect size.

6 Theory model (Section VI): “all the theoretical models” + core derivations

6.1 Setup and timeline

There is a continuum of risk-neutral entrepreneurs (borrowers) and a competitive lender. Time is $t = 0, 1, 2$:

- **Production stage** ($t = 0 \rightarrow 1$): the entrepreneur repeatedly operates a technology n times, producing outcomes $y_i \in \{0, \theta\}$.
- **Lending stage** ($t = 1 \rightarrow 2$): she requests an unsecured loan of size 1 and repays r if successful; defaults yield 0 to the lender.

6.2 Types and diversion risk

Entrepreneurs are either *good* (G) or *bad* (B), differing in the probability of “failure” (interpreted as diversion):

$$\Pr(y_i = \theta \mid j) = 1 - \pi_j, \quad \Pr(y_i = 0 \mid j) = \pi_j, \quad \pi_G < \pi_B.$$

Borrowers know their type; the lender does not.

6.3 Cash vs. cashless as a verifiability/precision choice (Equation (5))

At $t = 0$, the entrepreneur chooses a payment method:

- With **cash**, production outcomes are *unverifiable*.
- With **cashless**, each production outcome generates a verifiable record $x_i \in \{0, \theta\}$ with precision parameter $q \in [1/2, 1]$:

$$\begin{aligned} \Pr(x_i = \theta \mid y_i = \theta) &= \Pr(x_i = 0 \mid y_i = 0) = q, \\ \Pr(x_i = 0 \mid y_i = \theta) &= \Pr(x_i = \theta \mid y_i = 0) = 1 - q. \end{aligned} \tag{5}$$

Thus larger q means more informative records (mapping to more precise payment technologies and, empirically, to outflows and information-intensive payments).

6.4 Default cost for cashless borrowers (Equation (6))

If a borrower adopts cashless payments and then defaults, she suffers a cost:

$$C = \phi \sum_{i=1}^n (\theta - x_i), \quad \phi > 0. \quad (6)$$

Intuition: cashless records create a benchmark against capacity θ ; discrepancies in records upon default raise expected legal/reputational costs (“willful defaulter” channel, Indian context).

6.5 Borrower utilities (Equations (7)–(8))

If a borrower of type $j \in \{G, B\}$ takes the cashless contract (r_{cashless}, C) :

$$w_j(r_{\text{cashless}}, C \mid \pi_j) = (1 - \pi_j)(\theta - r_{\text{cashless}}) - \pi_j \mathbb{E}[C \mid \pi_j]. \quad (7)$$

If she takes the cash contract r_{cash} :

$$w_j(r_{\text{cash}} \mid \pi_j) = (1 - \pi_j)(\theta - r_{\text{cash}}). \quad (8)$$

6.6 Expected default cost (Equation (13)) — derivation sketch

From the cost definition (6), we need $\mathbb{E}[C \mid \pi_j]$. Since $C = \phi \sum (\theta - x_i)$ and outcomes are i.i.d.,

$$\mathbb{E}[C \mid \pi_j] = \phi n (\theta - \mathbb{E}[x_i \mid \pi_j]).$$

Now compute $\mathbb{E}[x_i \mid \pi_j]$:

$$\Pr(x_i = \theta \mid \pi_j) = \Pr(y_i = \theta)(q) + \Pr(y_i = 0)(1 - q) = (1 - \pi_j)q + \pi_j(1 - q),$$

so $\mathbb{E}[x_i \mid \pi_j] = \theta \Pr(x_i = \theta \mid \pi_j)$ and therefore

$$\mathbb{E}[C \mid \pi_j] = \phi n \theta (1 - q - \pi_j + 2q\pi_j). \quad (13)$$

Comparative statics: $\mathbb{E}[C \mid \pi_j]$ increases in n , ϕ , q , and π_j .

6.7 Competitive pricing and break-even (implied loan rates)

Because the lender is competitive and risk-neutral, it breaks even on each contract. If the lender believes the borrower has default probability π , then with unit principal the break-even condition is:

$$(1 - \pi)r = 1 \quad \Rightarrow \quad r(\pi) = \frac{1}{1 - \pi}.$$

In a separating equilibrium, the lender infers:

$$r_{\text{cashless}} = \frac{1}{1 - \pi_G}, \quad r_{\text{cash}} = \frac{1}{1 - \pi_B},$$

and under pooling,

$$r_{\text{pool}} = \frac{1}{1 - \bar{\pi}}, \quad \bar{\pi} = \alpha\pi_G + (1 - \alpha)\pi_B.$$

6.8 Incentive compatibility and individual rationality (Equations (9)–(12))

A separating equilibrium requires that each type chooses its intended contract and prefers participating:

$$(IC_G) \quad (1 - \pi_G)(\theta - r_{\text{cashless}}) - \pi_G \mathbb{E}[C \mid \pi_G] \geq (1 - \pi_G)(\theta - r_{\text{cash}}), \quad (9)$$

$$(IC_B) \quad (1 - \pi_B)(\theta - r_{\text{cash}}) \geq (1 - \pi_B)(\theta - r_{\text{cashless}}) - \pi_B \mathbb{E}[C \mid \pi_B], \quad (10)$$

$$(IR_G) \quad (1 - \pi_G)(\theta - r_{\text{cashless}}) - \pi_G \mathbb{E}[C \mid \pi_G] \geq 0, \quad (11)$$

$$(IR_B) \quad (1 - \pi_B)(\theta - r_{\text{cash}}) \geq 0. \quad (12)$$

6.9 Proposition 1 (selection effect): existence of separating equilibrium

Proposition 1 gives conditions under which good types choose cashless (to signal) while bad types stick with cash. Substituting break-even rates and (2) into IC constraints yields the paper's bounds on $\phi n \theta$:

$$\frac{\pi_B - \pi_G}{\pi_B(1 - \pi_G)(1 - q - \pi_B + 2q\pi_B)} \leq \phi n \theta \leq \frac{\pi_B - \pi_G}{(1 - \pi_B)\pi_G(1 - q - \pi_G + 2q\pi_G)}. \quad (14)$$

Interpretation: the expected default cost from adopting cashless payments must be *intermediate*:

- large enough to deter bad types (otherwise they mimic),
- not too large to deter good types (otherwise no one goes cashless).

The paper also requires that pooling is not too attractive (average quality not too high), expressed as:

$$\frac{1 - \bar{\pi}}{1} \leq \frac{1 - \pi_G}{1 + \phi n \theta \pi_G(1 - q - \pi_G + 2q\pi_G)}. \quad (15)$$

This is consistent with contexts where adverse selection is severe (e.g., subprime unsecured lending), so separation is valuable.

Comparative statics (link back to the empirics). A separating equilibrium is more likely (selection effect is stronger) when:

- n is larger: more cashless records $\Rightarrow$ stronger commitment / benchmark;
- q is larger: more precise payment technology $\Rightarrow$ stronger signaling.

This maps directly to the paper’s empirical evidence: stronger effects for information-intensive payments and for outflows (Table V).

6.10 Proposition 2 (accuracy effect): complementarity with borrower quality

Proposition 2 states:

$$\frac{d(r_{\text{cash}} - r_{\text{cashless}})}{d(\pi_B - \pi_G)} > 0. \quad (16)$$

Using $r(\pi) = 1/(1 - \pi)$, we have

$$r_{\text{cash}} - r_{\text{cashless}} = \frac{1}{1 - \pi_B} - \frac{1}{1 - \pi_G},$$

which is increasing in the “type gap” ($\pi_B - \pi_G$). Intuition: when types are more separated in risk, revealing oneself as good (via cashless adoption) buys a larger reduction in borrowing cost.

Empirical connection. This comparative static has an interaction-term interpretation: the benefit of cashless usage is larger for genuinely better borrowers. Table VII’s interactions with credit score and payment volatility align with this “accuracy effect.”

6.11 Appendix C: alternative information structure (robustness of the model)

The paper also considers an *asymmetric* information structure (inspired by He, Huang, and Zhou, 2023) in which cashless technologies disproportionately improve the detection of bad outcomes (“subprime” realizations). Under that alternative structure, the expected default cost for a type- j borrower becomes:

$$\mathbb{E}[C \mid \pi_j] = \phi n \theta q \pi_j, \quad (\text{C23})$$

which is monotone increasing in precision q for any $\pi_j > 0$. The key message is that the paper’s separating-equilibrium logic (selection effect) and the comparative statics (accuracy effect) continue to hold under this alternative formulation.